

Puglia Winery Case Study Report

Kelsey Greaves

Chike Osakwe

Nadia Rahbany

Elegant Tan

Bami Walker-Thomas

College of Business and Economics, California State University, Fullerton

ISDS 556: Data Warehousing and Foundations of Business Intelligence

Dr. Peng Liu

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Project Overview

Puglia Winery has experienced significant growth in wine production and sales over the past three years, expanding its customer base throughout California, the United States, and the United Kingdom. While this growth has created new business opportunities, it has also increased the complexity of managing production, inventory, and sales operations. Currently, management relies heavily on experience and intuition to make strategic decisions because business information is stored across separate operational systems that are difficult to analyze together.

The purpose of this project is to propose a Business Intelligence (BI) and Enterprise Data Warehouse (EDW) solution that integrates production and merchant sales data into a centralized repository. Following the Kimball Business Dimensional Lifecycle methodology, the proposed data warehouse will provide a single, reliable source of information that supports strategic decision-making through historical reporting, profitability analysis, and performance measurement. By consolidating data from multiple systems, Puglia Winery will be able to answer critical business questions regarding product profitability, customer value, and market performance using accurate and timely information.

Current Analytical Gaps and Organizational Readiness

Although Puglia Winery collects valuable operational data, its current analytical capabilities are limited because information is stored in two independent transactional systems. The production system uses an Oracle database to manage product and production information, while the merchant sales system uses Microsoft SQL Server to manage customer and sales transactions. Because these systems use different database platforms, data structures, and formats, management cannot easily combine information to produce meaningful reports or perform comprehensive business analysis.

As a result, several analytical gaps exist:

- Product profitability cannot be accurately measured because production costs and sales revenue are stored in separate systems.
- Customer profitability is difficult to evaluate since purchasing history, revenue, and production costs cannot be analyzed together.
- Market performance across California, the rest of the United States, and international markets cannot be compared efficiently.
- Historical trend analysis by month, season, or year requires significant manual effort and is prone to inconsistencies.
- Decision-making relies primarily on management experience rather than consistent, data-driven insights.

Despite these challenges, Puglia Winery appears well prepared to implement a Business Intelligence solution. The organization already captures the necessary production, product, customer, and sales data within its existing operational systems. Because the required business data is available, implementing an Enterprise Data

Warehouse will primarily involve integrating, cleansing, transforming, and organizing this information into a centralized analytical environment. This provides a strong foundation for supporting future business growth through improved reporting and decision support.

High-Level Enterprise Data Warehouse Bus Matrix

Table 1

Puglia Winery Data Warehouse Bus Matrix

Business Process (Fact Tables)	Date	Product	Customer	Market	Production Year
Production History	✓	✓			✓
Merchant Sales	✓	✓	✓	✓	
Product Profitability	✓	✓		✓	✓
Customer Profitability	✓	✓	✓	✓	✓
Market Performance	✓	✓		✓	

Consistent with Kimball's Enterprise Data Warehouse Bus Architecture, the proposed solution identifies the winery's primary business processes and the conformed dimensions that will be shared across analytical subject areas. The bus matrix provides a high-level roadmap for developing the enterprise data warehouse while ensuring consistency across future analytical applications. The proposed bus matrix identifies the major business processes that will populate the Enterprise Data Warehouse and the conformed dimensions shared across those processes. The Date and Product dimensions are common across nearly all analytical subject areas, while the Customer and Market dimensions support sales and profitability analysis. Integrating these dimensions enables Puglia Winery to analyze sales, production costs, profit margins, customer behavior, and market performance consistently across the organization. The bus architecture also provides scalability, allowing additional business processes and dimensions to be incorporated as the winery continues to expand.

Business Intelligence System Functionality and Features

The proposed Business Intelligence (BI) system will provide Puglia Winery with a centralized platform for reporting, analysis, and decision support. By integrating production and merchant sales data into an Enterprise Data Warehouse (EDW), the BI system will enable managers to monitor business performance through interactive dashboards, standard reports, and analytical queries. The system will replace manual reporting and intuition-based decision-making with accurate, timely, and data-driven insights.

The BI system will allow users to analyze information across multiple dimensions, including product, customer, market, and time. Users will also be able to drill down from summary information to detailed transaction data, compare performance across different time periods, and identify trends that support strategic planning.

The business problems that will be focused on for this data warehousing project are the profitability of products, the identification of key customers, and profitability of each market. Puglia Winery has a need for forecasting the customer demand of each product so production can plant additional grapes as demand necessitates. They also need to analyze the cost and margins for each market to determine which market they should focus on in the future. The key performance indicators for these processes are unit sales (cases/dozens), dollar sales, cost, and margin(revenue) for each product, customer, wine type, and market by month, season and year. These metrics will help management identify the most profitable wine products and determine which products should receive additional production or marketing investment. They will also help the sales manager identify high-value merchants, strengthen customer relationships, and develop targeted sales strategies. Finally, these measures will enable management to evaluate the profitability of California, the rest of the United States, and international markets while supporting future expansion decisions. A list of candidate reports is included below as Figure 1.

Each department, sales, production, and executive, require unique reports to be run on a regular basis. The sales department, managed by Stacie Giano, will analyze customer purchasing behavior, evaluate sales performance across markets, identify top-performing customers, and monitor revenue trends. The production department, managed by Tommy Siragusa, will review production volumes, production costs, and profitability by wine variety to support production planning and quality management.. Finally, Jack Gillespie, managing director, will use executive dashboards and strategic reports to evaluate profitability, monitor business performance, identify growth opportunities, and support long-term planning.

Figure 1
Candidate Reports

Doc title:		Project: WINEWISE				Prepared by: Group		Date Prepared: 7/29/2026	
#	Report name	Short Description	Report Category	Primary Owner	Bus. Value	Level of Effort	Report Type	Frequency	Data Elements
1	Total Sales / Revenue Product Ranking	total sales and revenue by product type for current year and prior year, with rank for each year and change in rank	Sales/Production	Stacie/Tommy	9	3	table	Monthly	year, product type, sales dollars, margin dollars
2	Total Sales Customer Ranking	total sales and revenue by customer for current year and prior year, with rank for each year and change in rank	Sales	Stacie	9	3	table	Monthly	year, sales dollars, margin dollars, customer
3	Total Sales Market Ranking	total sales and revenue by market for current year and prior year, with rank for each year and change in rank	Sales	Stacie	9	3	table	Monthly	year, sales dollars, margin dollars, customer, market
4	Seasonal Sales Trend Report	Sales performance by season and year	Executive	Jack	8	3	table and line chart	Quarterly	sales dollars, cost, margin, season, year, forecast dollars
5	Market Orders Time Series	last 12 months of actual orders vs forecast by market	Executive/Sales	Jack/Stacie	8	4	table and line chart	Monthly	month, quarter, season, market, sales dollars, forecast dollars, customer
6	Product Orders Time Series	last 12 months of actual orders vs forecast by product	Executive/Production	Jack/Tommy	9	4	table and line chart	Monthly	month, quarter, season, product, sales dollars, cost, forecast dollars, forecast orders
7	Customer Market Breakdown	last 12 months of customers by market	Executive	Jack	6	3	pie chart	Quarterly	month, customer, market, sales dollars, cost, margin
8	Revenue YTD and Forecast	total revenue by month for the last 12 months and forecasted revenue for the next three months	Executive	Jack	7	4	Matrix and line chart	Monthly	sales dollars, cost, margin, month, forecast dollars

BI Applications

The Enterprise Data Warehouse will provide data to Business Intelligence applications that present information through dashboards, reports, and interactive analysis tools. Recommended applications include:

- Executive dashboards for monitoring key business metrics.
- Interactive reporting tools that allow users to filter information by product, customer, market, and time.
- Online Analytical Processing (OLAP) capabilities for multidimensional analysis and drill-down reporting.
- Data visualization tools that display trends using charts, graphs, and performance indicators.
- Ad hoc query tools that enable managers to answer business questions without requiring technical support.

Reporting and Analysis Requirements

The BI system should support both operational monitoring and strategic decision-making through different reporting methods.

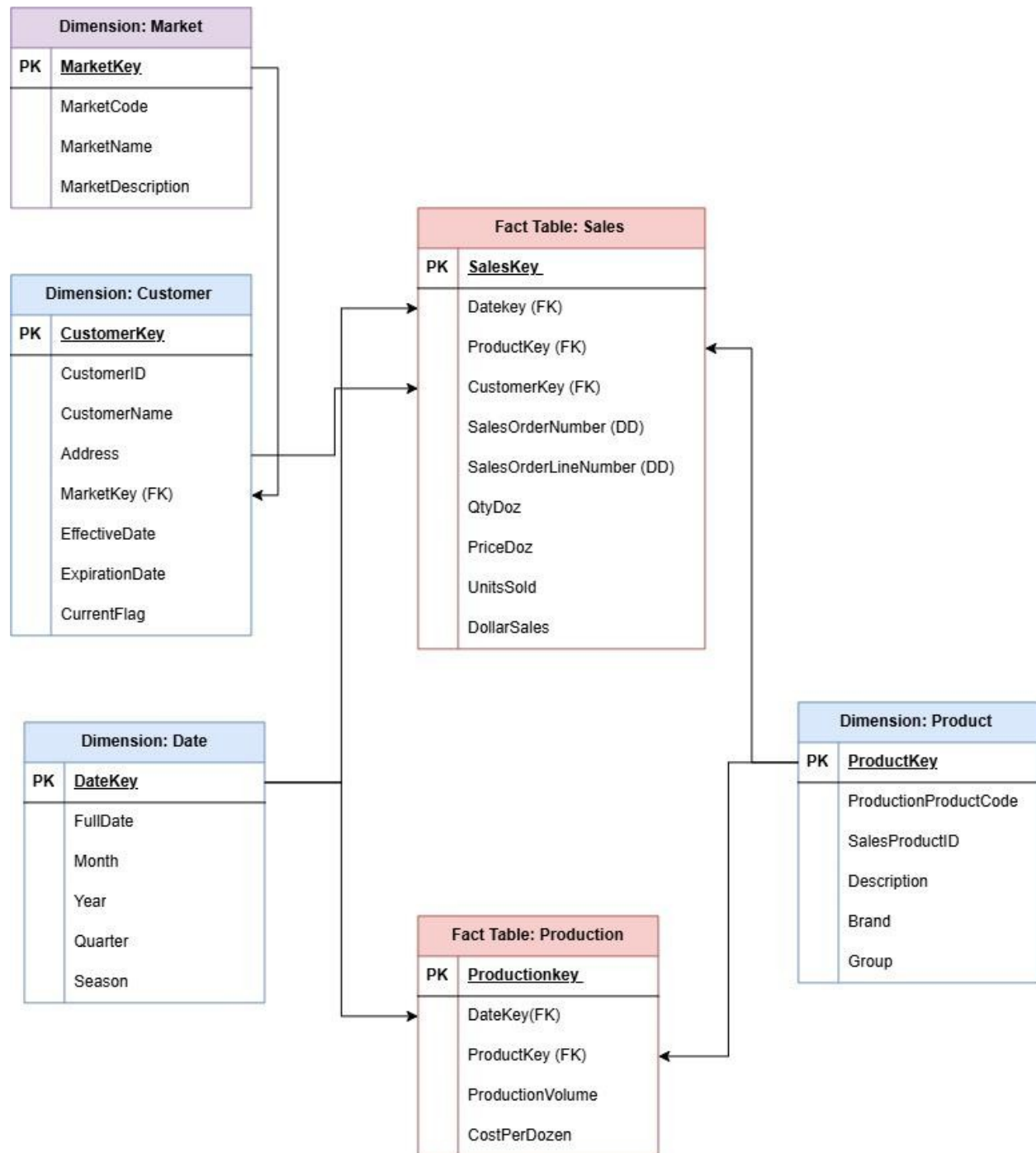
- **Daily or Weekly Dashboards:** Monitor current sales activity and key performance indicators.
- **Monthly Management Reports:** Evaluate profitability by product, customer, and market.
- **Quarterly Trend Analysis:** Compare seasonal performance and identify emerging business trends.
- **Annual Executive Reports:** Assess long-term performance, support production planning, and guide future investment decisions.

The system should also provide analytical capabilities such as drill-down, filtering, trend analysis, year-over-year comparisons, and ranking of products, customers, and markets. These features will enable management to quickly identify opportunities, address performance issues, and make informed business decisions based on reliable data.

Data Model

Puglia Winery seeks to identify the most profitable products, key customers, and markets, all of which depend on the same core business measures, including unit sales, dollar sales, cost, and margin. To support the analysis of these business metrics, the data model is designed using a Kimball-inspired galaxy schema. The model contains two fact tables because the winery has two distinct business processes: production and sales. The Fact Sales table captures individual sales transactions at the grain of one row per sales transaction line, while the Fact Production table records production results at the grain of one row per product per year. The schema also supports filtering and aggregating data across different time periods, products, customers, and markets, making it well-suited for the winery's analytical and reporting needs.

Figure 2
Data Model for Pulgia Winery Data Warehouse



As shown in Figure 2, the Market dimension is snowflaked from the Customer dimension. Snowflaking was applied because multiple customers can belong to the same market. Separating market information into its own dimension reduces data redundancy, improves data consistency, and simplifies maintenance when market

attributes change. The Product and Market dimensions are implemented as Type 1 slowly changing dimensions because updates to these attributes are expected to overwrite existing values rather than maintain historical versions. The Customer dimension is implemented as a Type 2 slowly changing dimension to preserve historical customer information, such as address changes, over time.

Data Analysis

The proposed business intelligence solution for Puglia Winery will integrate data from the organization's two existing operational systems into a centralized data warehouse. Currently, the winery stores production data in an Oracle-based production system and merchant sales data in a Microsoft SQL Server database. While these systems effectively support daily operations, their separate database platforms and structures make it difficult to analyze business performance across the organization. Integrating these data sources will provide management with a comprehensive view of product profitability, customer performance, and market trends.

The production system contains product information, product groups, production history, production volumes, and average production costs. The merchant sales system contains customer information, product information, sales orders, and sales order line details, including quantities sold and selling prices. Together, these systems provide the data necessary to measure sales revenue, production costs, profit margins, customer purchasing behavior, and market performance.

Both systems are primary data sources because the information is generated directly from the winery's daily business operations. These operational data are appropriate for supporting the organization's business intelligence objectives because they provide accurate information needed to answer the three business questions identified in the case study: determining the most profitable products, identifying key customers, and evaluating the profitability of different markets.

Although the case study does not indicate major data quality issues, moderate data cleansing will be necessary before the data are loaded into the warehouse. Since the two operational systems use different database management systems, product identifiers, customer information, date formats, and market classifications should be standardized to ensure consistency. Validation checks should also identify duplicate records, missing values, and invalid data before loading. These cleansing activities will improve data quality and ensure that reports generated from the warehouse are accurate and reliable.

Table 2*Comparison of Production System and Merchant Sales System*

Business Measure	Production System	Merchant Sales System
Production Volume	✓	
Production Cost	✓	
Unit Sales		✓
Sales Revenue		✓
Profit Margin	✓	✓
Product Profitability	✓	✓
Customer Profitability	✓	✓
Market Profitability		✓

Extraction, Transformation, and Loading

The ETL process will integrate data from the Oracle production system and the SQL Server merchant sales system into a centralized dimensional data warehouse. The source systems operate independently and use different database platforms; ETL is essential for creating consistent, accurate, and reliable information sources for business intelligence reports.

During the extraction phase, relevant data will be collected from both operational systems and transferred to a staging area. The staging environment provides a temporary workspace where data can be validated and transformed before entering the warehouse, preventing incomplete or inaccurate data from affecting analytical reports.

The transformation phase standardizes data from both systems by aligning product identifiers, customer information, date formats, and market classifications. Validation checks should ensure that sales transactions reference valid customers and products, required fields are complete, and duplicate or invalid records are identified before loading. Business measures such as sales revenue and profit margin should also be calculated during transformation so that reports utilize the same business definitions. Additionally, detailed transaction data will be preserved, while aggregated data by month, season, year, product, customer, and market will be maintained to support management reporting, allowing the information to be simpler to understand.

The dimensional model also introduces selective denormalization by organizing data into fact and dimension tables, which improves query performance compared to highly normalized operational databases. As the data warehouse grows, partitioning the fact table by time period, such as year, can further improve query performance and simplify ongoing maintenance.

After transformation, the cleaned data will be loaded into the dimensional warehouse. Dimension tables should be loaded before the fact table to maintain referential integrity, and historical data should be retained to support long-term trend analysis. Since management primarily analyzes monthly, seasonal, and annual performance, a nightly batch ETL process is recommended. Running the ETL process after business hours

minimizes the impact on operational systems while providing managers with updated information each business day. Real-time or near real-time loading is not necessary because the winery's strategic reporting requirements do not require immediate access to transactional data.

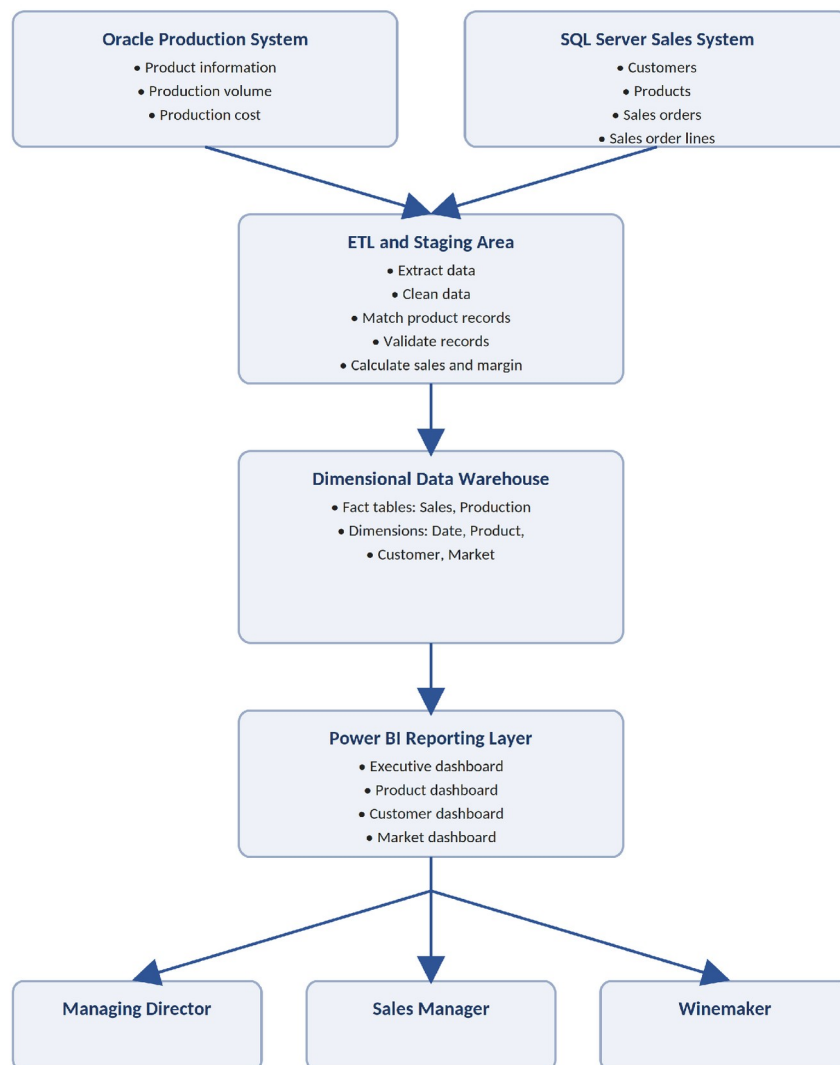
Architecture Recommendation

The recommended design is a centralized data warehouse that combines information from the Oracle production system and the SQL Server sales system.

The architecture includes both back-room and front-room technologies. The back-room handles data extraction, cleansing, storage, and processing. The front-room provides dashboards and reports for managers and other users.

Figure 3

Recommended Architecture Design for Pulgia Winery



Back-Room Technologies

The Oracle production system and SQL Server sales system will remain the official systems of record. The warehouse will only read data from them and will not make changes to the original systems. This reduces risk and allows employees to continue using the systems they already know.

SQL Server Integration Services, or SSIS, is recommended as the ETL tool. SSIS can connect to both Oracle and SQL Server and can handle data extraction, cleansing, matching, calculations, scheduling, and error handling.

Azure Data Factory could also be considered later if the winery decides to move more of its systems to the cloud. For now, SSIS is a practical choice because it works well with the proposed SQL Server warehouse.

The staging area will also be stored in SQL Server. It will hold:

- Raw source data
- Cleaned and standardized data
- Rejected records
- Audit information
- Product-matching tables

Because the Oracle production system and SQL Server sales system contain product data in different formats, a product crosswalk will be used to match the ProductionProductCode from Oracle with the SalesProductID from SQL Server. The product data will then be standardized before it is loaded into the shared Product dimension.

The staging area should not be directly available to business users.

Data Warehouse

The warehouse will be built in SQL Server. This is a good fit because the sales system already uses SQL Server, and it integrates naturally with SSIS and Power BI.

The warehouse will include two main fact tables.

Sales Fact Table

The sales fact table will contain:

- Date key
- Product key
- Customer key
- Sales order number
- Sales order line number
- Quantity dozen

- Price per dozen
- Dollar sales

The Customer dimension will connect to the Sales fact table, and the Market dimension will snowflake off of the Customer dimension.

Production Fact Table

The production fact table will contain:

- Date key
- Product key
- Production volume
- Cost per dozen

Because production results are recorded annually, the Date key will represent the applicable production year.

Both fact tables will share only the Date and Product dimensions. This will make it easier to compare sales and production using the same product and time definitions. The Customer dimension will connect only to the Sales fact table, while the Market dimension will snowflake off of the Customer dimension.

Front-Room Technologies

Power BI is recommended for reporting and dashboards. It can connect directly to the SQL Server warehouse and allow users to filter information by product, customer, market, month, season, and year.

The Power BI model should include standardized measures such as:

- Total quantity sold
- Total sales
- Total cost
- Total margin
- Margin percentage
- Average selling price
- Production volume
- Sales by product
- Sales by customer
- Sales by market

Using shared measures will help make sure that everyone is using the same calculations.

Recommended Dashboards

Executive Dashboard

The executive dashboard should give the managing director a high-level view of the winery's performance. It should include:

- Total sales
- Total cost
- Total margin
- Margin percentage
- Top products
- Top customers
- Best-performing markets
- Sales trends over time

Product Profitability Dashboard

This dashboard should show:

- Quantity sold by product
- Sales by product
- Cost by product
- Margin by product
- Production compared with sales
- Red wine compared with white wine
- Product trends over time

This report will help management identify the most profitable products.

Customer Dashboard

The customer dashboard should include:

- Sales and margin by merchant
- Products purchased by each customer
- Top and lowest-performing customers
- Customer trends over time
- Customer performance by market and season

This will help the winery identify its most valuable customers.

Market Dashboard

The market dashboard should compare:

- California

- Rest of the United States
- International sales

It should show sales volume, total sales, margin, market share, and monthly trends. This will help management determine which market is performing the best.

Security and Access

Access should be based on each employee's responsibilities.

User	Recommended access
Managing director	All dashboards and reports
Sales manager	Sales, customer, product, and market reports
Winemaker	Production, product, cost, and sales-volume reports
System administrator	ETL, database, security, and refresh management

Business users should have read-only access to reports. They should not be able to change warehouse data.

The service accounts used to extract data from Oracle and SQL Server should also have read-only access. Passwords and connection information should be stored securely.

Monitoring and Backups

The ETL process should keep an audit log that records:

- Start and end time
- Number of records extracted
- Number of records loaded
- Number of records rejected
- Whether the load succeeded or failed
- Details about any errors
- Number of product records that could not be matched

The system administrator should be alerted when a load fails, when an unusually large number of records are rejected, or when products cannot be matched between the two source systems.

The staging area, warehouse, ETL packages, and Power BI files should be backed up regularly. Daily backups and weekly full backups would be a reasonable starting point. The winery should also test the recovery process from time to time to make sure the backups work.

Rollout Plan

The project should be introduced in two stages.

The first stage will include:

- Building the staging area
- Creating the main dimensions
- Creating the product crosswalk
- Creating the sales fact table
- Developing the product, customer, and market dashboards

The second stage will include:

- Adding the production fact table
- Comparing production with sales
- Improving data-quality monitoring
- Adding forecasting or other advanced analysis

Overall, a SQL Server data warehouse supported by SSIS and Power BI would give Puglia Winery a practical way to bring its information together. It would also allow management to make decisions based on actual sales, production, cost, customer, product, and market data instead of relying mainly on experience or intuition.

Conclusion

This report proposed a Business Intelligence (BI) and Enterprise Data Warehouse (EDW) solution to address Puglia Winery's growing need for data-driven decision-making. The analysis identified that the winery's current Oracle production system and SQL Server merchant sales system operate independently, limiting management's ability to analyze product profitability, customer performance, and market trends. By integrating these systems into a centralized data warehouse using the Kimball Business Dimensional Lifecycle approach, Puglia Winery will gain a single, reliable source of information that supports accurate reporting and strategic analysis.

The proposed solution includes an Enterprise Data Warehouse with shared dimensions and fact tables, an ETL process to extract, transform, and load data from both operational systems, and Power BI dashboards that provide executives, sales managers, and production managers with timely and meaningful business insights. Standardized KPIs, interactive reports, and historical trend analysis will enable management to identify profitable products, strengthen relationships with key customers, evaluate market performance, and make informed decisions based on consistent and reliable data rather than intuition.

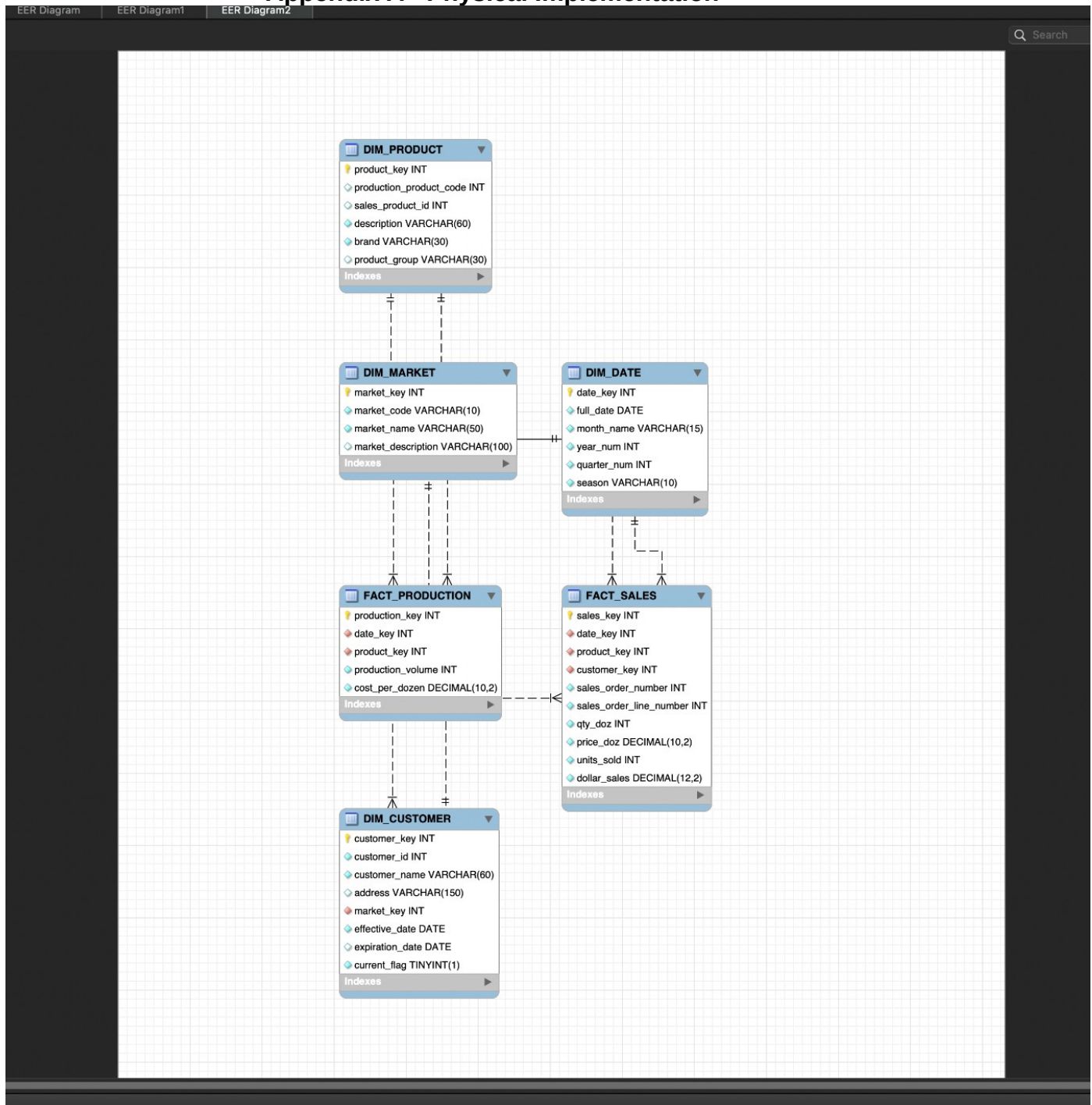
Next Steps

To successfully implement the proposed solution, Puglia Winery should proceed with the following steps:

1. Validate the business requirements and reporting needs with key stakeholders, including the Managing Director, Sales Manager, and Winemaker.
2. Design and implement the dimensional data warehouse, including the conformed dimensions and fact tables identified in the Enterprise Bus Matrix.
3. Develop and test the ETL process to integrate data from the Oracle production system and SQL Server merchant sales system while ensuring data quality and consistency.
4. Build the recommended Power BI dashboards and management reports, incorporating the defined KPIs for products, customers, and markets.
5. Conduct user acceptance testing and provide training so managers can effectively use the BI system for reporting and analysis.
6. Deploy the solution using the phased rollout plan, beginning with sales reporting and dashboards before expanding to production analytics and more advanced forecasting capabilities.
7. Establish ongoing maintenance, monitoring, backups, and data governance procedures to ensure the long-term reliability, security, and scalability of the Business Intelligence environment.

By following these recommendations, Puglia Winery will be well-positioned to support future growth with an integrated Business Intelligence solution that improves operational visibility, enhances strategic planning, and enables informed, data-driven business decisions.

Appendix A - Physical Implementation



Appendix B- SQL Code

```
DROP DATABASE IF EXISTS puglia_winery_dw;
```

```

CREATE DATABASE puglia_winery_dw;

USE puglia_winery_dw;

CREATE TABLE DIM_MARKET (
    market_key INT AUTO_INCREMENT PRIMARY KEY,
    market_code VARCHAR(10) NOT NULL,
    market_name VARCHAR(50) NOT NULL,
    market_description VARCHAR(100)
) ENGINE = InnoDB;

CREATE TABLE DIM_CUSTOMER (
    customer_key INT AUTO_INCREMENT PRIMARY KEY,
    customer_id INT NOT NULL,
    customer_name VARCHAR(60) NOT NULL,
    address VARCHAR(150),
    market_key INT NOT NULL,
    effective_date DATE NOT NULL,
    expiration_date DATE,
    current_flag TINYINT(1) NOT NULL DEFAULT 1,
    CONSTRAINT fk_customer_market
        FOREIGN KEY (market_key)
        REFERENCES DIM_MARKET (market_key)
) ENGINE = InnoDB;

CREATE TABLE DIM_DATE (
    date_key INT PRIMARY KEY,
    full_date DATE NOT NULL,
    month_name VARCHAR(15) NOT NULL,
    year_num INT NOT NULL,
    quarter_num INT NOT NULL,
    season VARCHAR(10) NOT NULL
) ENGINE = InnoDB;

CREATE TABLE DIM_PRODUCT (
    product_key INT AUTO_INCREMENT PRIMARY KEY,
    production_product_code INT,
    sales_product_id INT,
    description VARCHAR(60) NOT NULL,
    brand VARCHAR(30) NOT NULL,
    product_group VARCHAR(30)
) ENGINE = InnoDB;

CREATE TABLE FACT_SALES (

```

```

sales_key INT AUTO_INCREMENT PRIMARY KEY,
date_key INT NOT NULL,
product_key INT NOT NULL,
customer_key INT NOT NULL,
sales_order_number INT NOT NULL,
sales_order_line_number INT NOT NULL,
qty_dozen INT NOT NULL,
price_dozen DECIMAL(10,2) NOT NULL,
units_sold INT NOT NULL,
dollar_sales DECIMAL(12,2) NOT NULL,
CONSTRAINT fk_sales_date
    FOREIGN KEY (date_key)
    REFERENCES DIM_DATE (date_key),
CONSTRAINT fk_sales_product
    FOREIGN KEY (product_key)
    REFERENCES DIM_PRODUCT (product_key),
CONSTRAINT fk_sales_customer
    FOREIGN KEY (customer_key)
    REFERENCES DIM_CUSTOMER (customer_key)
) ENGINE = InnoDB;

CREATE TABLE FACT_PRODUCTION (
    production_key INT AUTO_INCREMENT PRIMARY KEY,
    date_key INT NOT NULL,
    product_key INT NOT NULL,
    production_volume INT NOT NULL,
    cost_per_dozen DECIMAL(10,2) NOT NULL,
    CONSTRAINT fk_production_date
        FOREIGN KEY (date_key)
        REFERENCES DIM_DATE (date_key),
    CONSTRAINT fk_production_product
        FOREIGN KEY (product_key)
        REFERENCES DIM_PRODUCT (product_key)
) ENGINE = InnoDB;

INSERT INTO DIM_MARKET (
    market_key,
    market_code,
    market_name,
    market_description
)
VALUES
    (1, 'CA', 'California', 'California'),
    (2, 'US', 'Rest of United States', 'Rest of United States'),
    (3, 'INT', 'International', 'International');

```

```

INSERT INTO DIM_CUSTOMER (
    customer_key,
    customer_id,
    customer_name,
    address,
    market_key,
    effective_date,
    expiration_date,
    current_flag
)
VALUES
    (1, 1, 'Tasty Wines', '205 Soscol Ave, Napa, CA 94559-0400, USA',
1, '2020-01-01', NULL, 1),
    (2, 2, 'Cali Wines', '4000 Bel Aire Plaza, Napa, CA 94558-2835,
USA', 1, '2020-01-01', NULL, 1),
    (3, 3, 'London Wines', 'Queen Avenue, The Stratford, London, SW1A
1LZ, UK', 3, '2020-01-01', NULL, 1),
    (4, 4, 'The Sussex Wine Company', 'Wolf Road, Chichester, West
Sussex, PO20 7DU, UK', 3, '2020-01-01', NULL, 1),
    (5, 5, 'Merchant's Corner', '2241 Willow Rd, Glenview, IL 60025-
7636, USA', 2, '2020-01-01', NULL, 1),
    (6, 6, 'US Wines Direct', '115 N Randall Rd, Batavia, IL 60510-
9209, USA', 2, '2020-01-01', NULL, 1);

```

```

INSERT INTO DIM_DATE (
    date_key,
    full_date,
    month_name,
    year_num,
    quarter_num,
    season
)
VALUES
    (20200101, '2020-01-01', 'January', 2020, 1, 'Winter'),
    (20200201, '2020-02-01', 'February', 2020, 1, 'Winter'),
    (20200401, '2020-04-01', 'April', 2020, 2, 'Spring'),
    (20200701, '2020-07-01', 'July', 2020, 3, 'Summer'),
    (20201001, '2020-10-01', 'October', 2020, 4, 'Autumn'),
    (20210101, '2021-01-01', 'January', 2021, 1, 'Winter');

```

```

INSERT INTO DIM_PRODUCT (
    product_key,
    production_product_code,
    sales_product_id,

```

```

        description,
        brand,
        product_group
    )
VALUES
    (1, 1, 1, 'Bellarine Pinot Grigio', 'Bellarine', 'Pinot Grigio'),
    (2, 2, 2, 'Bellarine Pinot Noir', 'Bellarine', 'Pinot Noir'),
    (3, 3, 3, 'Downunder Merlot', 'Downunder', 'Merlot'),
    (4, 4, 4, 'Downunder Pinot Grigio', 'Downunder', 'Pinot Grigio'),
    (5, 5, 5, 'Downunder Pinot Noir', 'Downunder', 'Pinot Noir'),
    (6, 6, 6, 'Overhill Merlot', 'Overhill', 'Merlot'),
    (7, 7, 7, 'Overhill Pinot Noir', 'Overhill', 'Pinot Noir');

```

```

INSERT INTO FACT_SALES (
    sales_key,
    date_key,
    product_key,
    customer_key,
    sales_order_number,
    sales_order_line_number,
    qty_dozen,
    price_dozen,
    units_sold,
    dollar_sales
)

```

```

VALUES
    (1, 20200201, 1, 2, 227, 1, 50, 160.00, 600, 8000.00),
    (2, 20200201, 1, 3, 228, 1, 82, 110.00, 984, 9020.00),
    (3, 20200201, 2, 3, 229, 1, 29, 110.00, 348, 3190.00),
    (4, 20200201, 1, 5, 230, 1, 69, 100.00, 828, 6900.00),
    (5, 20200201, 1, 6, 231, 1, 96, 150.00, 1152, 14400.00),
    (6, 20200201, 3, 6, 231, 2, 20, 94.00, 240, 1880.00),
    (7, 20200201, 4, 1, 232, 1, 15, 100.00, 180, 1500.00);

```

```

INSERT INTO FACT_PRODUCTION (
    production_key,
    date_key,
    product_key,
    production_volume,
    cost_per_dozen
)

```

```

VALUES
    (1, 20200101, 1, 1120, 80.00),
    (2, 20200101, 2, 1090, 45.00),
    (3, 20200101, 3, 1349, 65.00),

```

```
(4, 20200101, 4, 423, 41.00),  
(5, 20200101, 5, 1422, 60.00),  
(6, 20200101, 6, 1187, 58.00),  
(7, 20200101, 7, 700, 50.00);
```

```
SELECT  
    table_name AS child_table,  
    column_name AS foreign_key,  
    referenced_table_name AS parent_table,  
    referenced_column_name AS parent_key  
FROM information_schema.key_column_usage  
WHERE table_schema = 'puglia_winery_dw'  
    AND referenced_table_name IS NOT NULL  
ORDER BY table_name, constraint_name;
```